

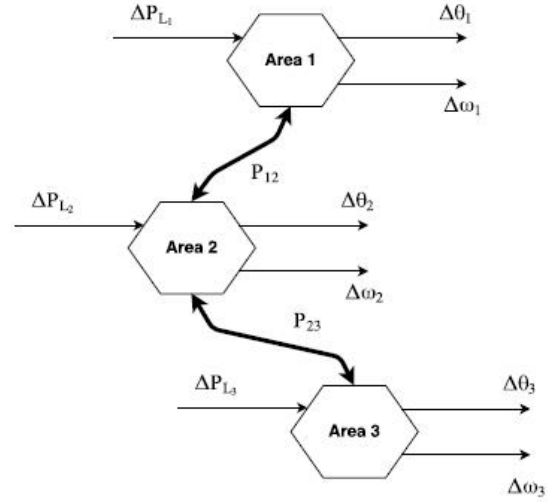
## Power Network System

We consider a Power Network System (PNS) composed of several power generation areas coupled through tie-lines. Tie lines allow for power exchange between areas. In this way, if the load profile of an area grows too much with respect to the power generated in that area, neighbouring areas can contribute to satisfy the request.

We consider thermal power stations with single-stage turbines. In the figure below the typical structure of a simple network of three areas is shown as an example along with measurable input and output variables. The term  $P_{ij}$  is function of the coupling between areas  $i$  and  $j$ .



(a) High voltage pylon.



(b) PNS structure.

Here we consider the dynamics of a PNS with five areas, where each area is endowed with primary control. The models are linearized around an equilibrium value. The state of area  $i$  consists of the vector  $x_i = [\Delta\theta_i, \Delta\omega_i, \Delta P_{m,i}, \Delta P_{v,i}]^T$  (all variables are assumed measurable for simplicity), where:

- $\Delta\theta_i$ : Deviation of the angular displacement of the rotor with respect to the stationary reference axis on the stator.
- $\Delta\omega_i$ : Speed deviation of rotating mass from nominal value.
- $\Delta P_{m,i}$ : Deviation of the mechanical power from nominal value (p.u.).
- $\Delta P_{v,i}$ : Deviation of the steam valve position from nominal value (p.u.).

The input  $u_i$  of each area is the deviation of the power set-point from the nominal value. A system composed of 5 areas, neglecting the disturbance term (i.e., the deviation of the requested load from the nominal value), can be described by the following model:

$$\dot{x} = Ax + Bu$$

where  $x = [x_1^T, \dots, x_5^T]^T$ ,  $u = [u_1, \dots, u_5]^T$ , and where the system matrices are specified in the file.

**Problem:**

1. Decompose the state and input vectors into subvectors, consistently with the physical description of the system. Obtain the corresponding decomposed model.
2. Generate the system matrices (both continuous-time and discrete-time, the latter with a sampling time  $h = 1$  s). Perform the following analysis:
  - a. Compute the eigenvalues and the spectral abscissa of the (continuous-time) system. Is it open-loop asymptotically stable?
  - b. Compute the eigenvalues and the spectral radius of the (discrete-time) system. Is it open-loop asymptotically stable?
3. For different state-feedback control structures (i.e., centralized, decentralized, and different distributed schemes) perform the following actions
  - a. Compute the continuous-time fixed modes
  - b. Compute the discrete-time fixed modes
  - c. Compute, if possible, the CONTINUOUS-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
  - d. Compute, if possible, the DISCRETE-TIME control gains using LMIs to achieve the desired performances. Apply, for better comparison, different criteria for computing the control laws.
  - e. Analyze the properties of the so-obtained closed-loop systems (e.g., stability, eigenvalues) and compute the closed-loop system trajectories (generated both in continuous-time and in discrete-time) of the speed deviations  $\Delta\omega_i$  starting from a common random initial condition.